

VINCENT AURELLIO BUDianto

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EDUCATION

National University of Singapore (NUS)

Aug 2021 – May 2025

Bachelor of Science (Honours)

- GPA: 4.5/5.0 | Dean's List AY2024/2025 Semester 1
- Major in Data Science and Analytics with specialization in Operations Research
- Minor in Quantitative Finance and Minor in Artificial Intelligence
- Relevant Courses: Machine Learning, Stochastic Processes, Data Visualization, Regression Analysis, Database Management, Quantitative Finance Fundamentals, AI Planning & Decision Making, Optimization, Big Data Systems

WORK EXPERIENCE

Junior AI Engineer, BlackFrost AI, Singapore, Singapore

Oct 2025 – Dec 2025

- Built and launched the first version of a business-facing AI chatbot platform from scratch using React (TypeScript), TailwindCSS, Redux, and Axios, supporting client-specific workflows and commercial deployment.
- Automated document-based workflows and company-specific processes via RAG pipelines on AWS, transforming previously manual tasks into AI-driven solutions and improving operational efficiency.
- Developed prompt orchestration and model workflow management to customize foundation models for each client, ensuring reliable, accurate, and context-aware responses.
- Implemented CI/CD pipelines and Dockerized the app, reducing setup time and enabling reliable deployments.

Data Analyst Intern, United Overseas Bank, Singapore, Singapore

May 2024 – Aug 2024

- Ensured data integrity in a large-scale data warehouse migration involving data from 5 countries, validating mappings using Hive & Spark SQL, and identifying critical mapping discrepancies to improve deployment efficiency.
- Optimized SQL queries from Hive to Spark, leveraging Spark's caching capabilities and query optimizations to reduce execution time by 40%, significantly boosting operational efficiency and scalability.
- Analyzed data for 2,000+ companies, identifying 50+ high-potential investment targets to support due diligence.
- Validated vendor and market data in Microsoft Excel, identifying discrepancies to improve procurement decisions.

PROJECTS

Impact Analysis of Adversarial Agent in Reinforcement Learning

Sep 2024 – Nov 2024

- Designed a dynamic grid world environment with stochastic wall placements and adversarial interference to evaluate agent robustness under varying conditions.
- Implemented PPO from scratch in Python, improving task success by ~25% under adversarial interference.

AI Chat Survey Chatbot

Feb 2024 – May 2024

- Collaborated in a team of 8 to develop an NLP-powered survey chatbot aimed at increasing engagement through personalized, conversational surveys as an alternative to traditional forms.
- Designed and implemented the chat interface using React TypeScript, integrated API calls for real-time messaging, and resolved critical frontend bugs such as duplicate message handling to ensure a smooth user experience.

TECHNICAL SKILLS

- **Programming:** Python, SQL, TypeScript, Java, R
- **ML & Data:** PyTorch, TensorFlow, Scikit-learn, JAX, Pandas, NumPy
- **Data & Cloud / DevOps:** AWS, Spark, Hive, Power BI, Docker, Git, GitHub Actions
- **Frontend & Web:** React, Redux, TailwindCSS, HTML/CSS, Axios

LEADERSHIP & ACTIVITIES

NUS Student Union Committee for Information Technology, NUS, Singapore

Training Director

May 2023 – Jun 2024

- Designed and led 20+ workshops on Python, Power BI, and Excel for 30+ students, ensuring hands-on learning.

Cyberia 2024 Project Director

Nov 2023 – Jan 2024

- Managed budgeting, scheduling, and stakeholder coordination to successfully execute the event on time and within scope while fostering teamwork through effective delegation and motivation enhancing productivity and engagement.

ADDITIONAL INFORMATION

- Developed portfolio website using HTML, showcasing data science projects using Python and R.